

Chapter 8

Discrete Probability

Discrete Structures for Computing

Discrete Probability

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Course outcomes



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Course learning outcomes

L.O.1	Understanding of logic and discrete structures
	L.O.1.1 – Describe definition of propositional and predicate logic
	L.O.1.2 – Define basic discrete structures: set, mapping, graphs
L.O.2	Represent and model practical problems with discrete structures
	L.O.2.1 – Logically describe some problems arising in Computing
	L.O.2.2 – Use proving methods: direct, contrapositive, induction
	L.O.2.3 – Explain problem modeling using discrete structures
L.O.3	Understanding of basic probability and random variables
	L.O.3.1 – Define basic probability theory
	L.O.3.2 – Explain discrete random variables
L.O.4	Compute quantities of discrete structures and probabilities
	L.O.4.1 – Operate (compute/ optimize) on discrete structures
	L.O.4.2 – Compute probabilities of various events, conditional ones, Bayes theorem

Motivations

- Gambling



- Real life problems



- **Computer Science:** cryptology – deals with encrypting codes or the design of error correcting codes





Which of these are **random phenomena**?

- The number you receive when rolling a **fair** dice
- The sequence for lottery special prize (by law!)
- Your blood type **(No!)**
- You met the red light on the way to school
 - The traffic light is **not** random. It has timer.
 - The pattern of **your riding** is random.

So what is special about randomness?

In the **long run**, they are predictable and have **relative frequency** (fraction of times that the event occurs over and over and over).



- **Experiment** (*thí nghiệm*): a procedure that yields one of a given set of possible outcomes.
 - Tossing a coin to see the face
- **Sample space** (*không gian mẫu*): set of possible **outcomes**
 - {Head, Tail}
- **Event** (*sự kiện*): a subset of sample space.
 - You see Head after an experiment. {Head} is an event.



Example

Example (1)

Experiment: Rolling a die. What is the sample space?

Answer: $\{1, 2, 3, 4, 5, 6\}$

Example (2)

Experiment: Rolling two dice. What is the sample space?

Answer: It depends on what we're going to ask!

- The total number?
 $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
- The number of each die?
 $\{(1,1), (1,2), (1,3), \dots, (6,6)\}$

Which is better?

The latter one, because they are **equally likely outcomes**



The Law of Large Numbers

Definition

The Law of Large Numbers (*Luật số lớn*) states that the **long-run relative frequency** of repeated independent events gets closer and closer to the **true** relative frequency as the number of trials increases.

Example

Do you believe that the true relative frequency of **Head** when you toss a coin is 50%?

Let's try!



Be Careful!

Don't misunderstand the Law of Large Numbers (LLN). It can lead to money lost and poor business decisions.

Example

I had 8 children, all of them are girls. Thanks to LLN (!?), there are high possibility that the next one will be a boy.
(Overpopulation!!!)

Example

I'm playing *Bầu cua tôm cá*, the fish has not appeared in recent 5 games, it will be more likely to be fish next game. Thus, I bet all my money in fish. **(Sorry, you lose!)**





Definition

The **probability** (*xác suất*) of an event E of a finite nonempty sample space of **equally likely outcomes** S is:

$$p(E) = \frac{|E|}{|S|}.$$

- Note that $E \subseteq S$ so $0 \leq |E| \leq |S|$
- $0 \leq p(E) \leq 1$
 - 0 indicates impossibility
 - 1 indicates certainty

People often say: "It has a **20%** probability"

Example (1)

What is the probability of getting a Head when tossing a coin?

Answer:

- There are $|S| = 2$ possible outcomes
- Getting a Head is $|E| = 1$ outcome, so $p(E) = 1/2 = 0.5 = 50\%$

Example (2)

What is the probability of getting a 7 by rolling two dice?

Answer:

- **Product rule:** There are a total of 36 equally likely possible outcomes
- There are six successful outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)
- Thus, $|E| = 6$, $|S| = 36$, $p(E) = 6/36 = 1/6$



Examples

Example (3)

We toss a coin 6 times. What is probability of H in 6th toss, if all the previous 5 are T?

Answer:

Don't be silly! Still $1/2$.

Example (4)

Which is more likely:

- Rolling an 8 when 2 dice are rolled?
- Rolling an 8 when 3 dice are rolled?

Answer:

Two dice: $5/36 \approx 0.139$

Three dice: $21/216 \approx 0.097$



Rule 1

A probability is a number **between 0 and 1**.

$$0 \leq p(E) \leq 1$$

Rule 2: Something has to happen rule

The probability of the set of all possible outcomes of a trial **must be 1**.

$$p(S) = 1$$

Rule 3: Compliment Rule

The probability of an event occurring is 1 minus the probability that it doesn't occur.

$$p(E) = 1 - p(\overline{E})$$





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Example

What is the probability of **NOT** drawing a heart card from 52 deck cards?

Answer:

Let E be the event of **getting** a heart from 52 deck cards. We have:

$$p(E) = 13/52 = 1/4$$

By the compliment rule, the probability of NOT getting a heart card is:

$$p(\overline{E}) = 1 - p(E) = 3/4$$



General Addition Rule

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

- If $E_1 \cap E_2 = \emptyset$: They are **disjoint**, which means they can't occur together
- then, $p(E_1 \cup E_2) = p(E_1) + p(E_2)$

Example

Example (1)

If you choose a number between 1 and 100, what is the probability that it is divisible by either 2 or 5?

Short Answer:

$$\frac{50}{100} + \frac{20}{100} - \frac{10}{100} = \frac{3}{5}$$

Example (2)

There are a survey that about 45% of VN population has Type O blood, 40% type A, 11% type B and the rest type AB. What is the probability that a blood donor has Type A or Type B?

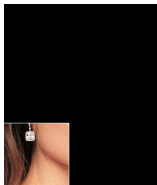
Short Answer:

$$40\% + 11\% = 51\%$$



Conditional Probability (Xác suất có điều kiện)

- “Knowledge” changes probabilities



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Definition

$p(E \mid F)$ = Probability of event E given that event F **has occurred**.

General Multiplication Rule

$$\begin{aligned} p(E \cap F) &= p(E) \times p(F \mid E) \\ &= p(F) \times p(E \mid F) \end{aligned}$$

Example

Example

What is the probability of drawing a red card and then another red card **without replacement** (*không hoàn lại*)?

Solution

E : the event of drawing the first red card

F : the event of drawing the second red card

$$p(E) = 26/52 = 1/2$$

$$p(F | E) = 25/51$$

So the event of drawing a red card and then another red card is

$$p(E \cap F) = p(E) \times p(F | E) = 1/2 \times 25/51 = 25/102$$



Definition

Events E and F are independent (*độc lập*) whenever

$$p(E | F) = p(E)$$

- The outcome of one event does not influence the probability of the other.
- Example: $p(\text{"Head"} | \text{"It's raining outside"}) = p(\text{"Head"})$
- If E and F are independent

$$p(E \cap F) = p(E) \times p(F)$$

Disjoint $\neq$ Independence

Disjoint events cannot be independent. They have no outcomes in common, so knowing that one occurred means the other did not.





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Example

If we know that the probability that a person has tuberculosis (TB) is $p(\text{TB}) = 0.0005$.

We also know $p(+|\text{TB}) = 0.999$ and $p(-|\overline{\text{TB}}) = 0.99$.

What is $p(\text{TB}|+)$ and $p(\overline{\text{TB}}|-)$?

Theorem (Bayes's Theorem)

$$p(F | E) = \frac{p(E | F)p(F)}{p(E | F)p(F) + p(E | \overline{F})p(\overline{F})}$$

Expected Value: Center

An insurance company charges \$50 a year. Can company make a profit? Assuming that it made a research on 1000 people and have following table:

Outcome	Payroll x	Probability $p(X = x)$
Death	10,000	$\frac{1}{1000}$
Disability	5000	$\frac{2}{1000}$
Neither	0	$\frac{997}{1000}$

- X is a **discrete random variable** (*biến ngẫu nhiên rời rạc*)

The company **expects** that they have to pay each customer:

$$E(X) = \$10,000\left(\frac{1}{1000}\right) + \$5000\left(\frac{2}{1000}\right) + \$0\left(\frac{997}{1000}\right) = \$20$$

Expected value (giá trị kỳ vọng)

$$E(X) = \sum x \cdot p(X = x)$$



Variance: The Spread

- Of course, the expected value \$20 will not happen in reality
- There will be **variability**. Let's calculate!

- Variance (*phương sai*)

$$V(X) = \sum (x - E(X))^2 \cdot p(X = x)$$

- $V(X) = 9980^2 \left(\frac{1}{1000}\right) + 4980^2 \left(\frac{2}{1000}\right) + (-20)^2 \left(\frac{997}{1000}\right) = 149,600$

- Standard deviation (*độ lệch chuẩn*)

$$SD(X) = \sqrt{V(X)}$$

- $SD(X) = \sqrt{149,600} \approx \386.78

Comment

The company expects to pay out \$20, and make \$30. However, the standard deviation of \$386.78 indicates that it's no sure thing. That's pretty big spread (and risk) for an average profit of \$20.





Example

Some people madly drink Coca-Cola, hoping to find a ticket to see Big Bang. Let's call tearing a bottle's label **trial** (*phép thử*):

- There are only possible outcomes (**congrats** or **good luck**)
- The probability of success, p , is the same on every trial, say 0.06
- The trials are independent. Finding a ticket in the first bottle does not change what might happen in the second one.

- **Bernoulli Trials**

- Another examples: tossing a coin many times, results of testing TB on many patients, ...

Geometric Model (Mô hình hình học)



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Question: How long it will take us to achieve a success, given p , the probability of success?

Definition (Geometric probability model: $\text{Geom}(p)$)

p = probability of success ($q = 1 - p$ = probability of failure)

X = number of trials until the first success occurs

$$p(X = x) = q^{x-1}p$$

Expected value: $\mu = \frac{1}{p}$

Standard deviation: $\sigma = \sqrt{\frac{q}{p^2}}$

Geometric Model: Example

Example

If the probability of finding a Sound Fest ticket is $p = 0.06$, how many bottles do you expect to open before you find a ticket? What is the probability that the first ticket is in one of the first four bottles?

Solution

Let X = number of trials until a ticket is found

We can model X with $\text{Geom}(0.06)$.

$$E(X) = \frac{1}{0.06} \approx 16.7$$

$$\begin{aligned} P(X \leq 4) &= P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\ &= (0.06) + (0.94)(0.06) + (0.94)^2(0.06) \\ &\quad + (0.94)^3(0.06) \\ &\approx 0.2193 \end{aligned}$$

Conclusion: *We expect to open 16.7 bottles to find a ticket. About 22% of time we'll find one within the first 4 bottles.*



Binomial Model (Mô hình nhị thức)

Previous Question: How long it will take us to achieve a success, given p , the probability of success?

New Question: You buy 5 Coca-Cola. What's the probability you get **exactly** 2 Sound Fest tickets?

Definition (Binomial probability model: $\text{Binom}(n, p)$)

n = number of trials

p = probability of success ($q = 1 - p$ = probability of failure)

X = number of successes in n trials

$$p(X = x) = \binom{n}{x} p^x q^{n-x}$$

Expected value: $\mu = np$

Standard deviation: $\sigma = \sqrt{npq}$



Binomial Model: Example

Example

Suppose you buy 20 Coca-Cola bottles. What are the mean and standard deviation of the number of winning bottles among them? What is the probability that there are 2 or 3 tickets?

Solution

Let X = number of tickets among $n = 20$ bottles

We can model X with $\text{Binom}(20, 0.06)$.

$$E(X) = np = 20(0.06) = 1.2$$

$$SD(X) = \sqrt{npq} = \sqrt{20(0.06)(0.94)} \approx 1.06$$

$$\begin{aligned} P(X = 2 \text{ or } 3) &= P(X = 2) + P(X = 3) \\ &= \binom{20}{2}(0.06)^2(0.94)^{18} + \binom{20}{3}(0.06)^3(0.94)^{17} \\ &\approx 0.2246 + 0.0860 = 0.3106 \end{aligned}$$

Conclusion: *In 20 bottles, we expect to find an average of 1.2 tickets, with a sd of 1.06. About 31% of the time we'll find 2 or 3 tickets among 20 bottles.*

